

WASP-11b

R-1405

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_191107 · created 2026-08-01T04:34:57+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458794.859 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.093 +/- 0.039
Transit depth (Rp/Rs)^2	0.86 +/- 0.71 [%]
Orbital Inclination (inc)	89.8 +/- 1.82
Airmass coefficient 1 (a1)	0.945 +/- 0.02
Airmass coefficient 2 (a2)	0.051 +/- 0.034
Scatter in the residuals of the lightcurve fit is	2.16 %
Best Comparison Star	#3 - [374, 310]
Optimal Method	PSF photometry
Transit Duration (day)	0.096 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.093 ± 0.039 vs 0.131 (deviation 0.88σ)
Duration fit vs published	0.096 d vs 0.1075 d (deviation 0.8σ)
Mid-time O–C	6.0 min
Depth SNR	2.2
Residual scatter	2.16 %

Light curve

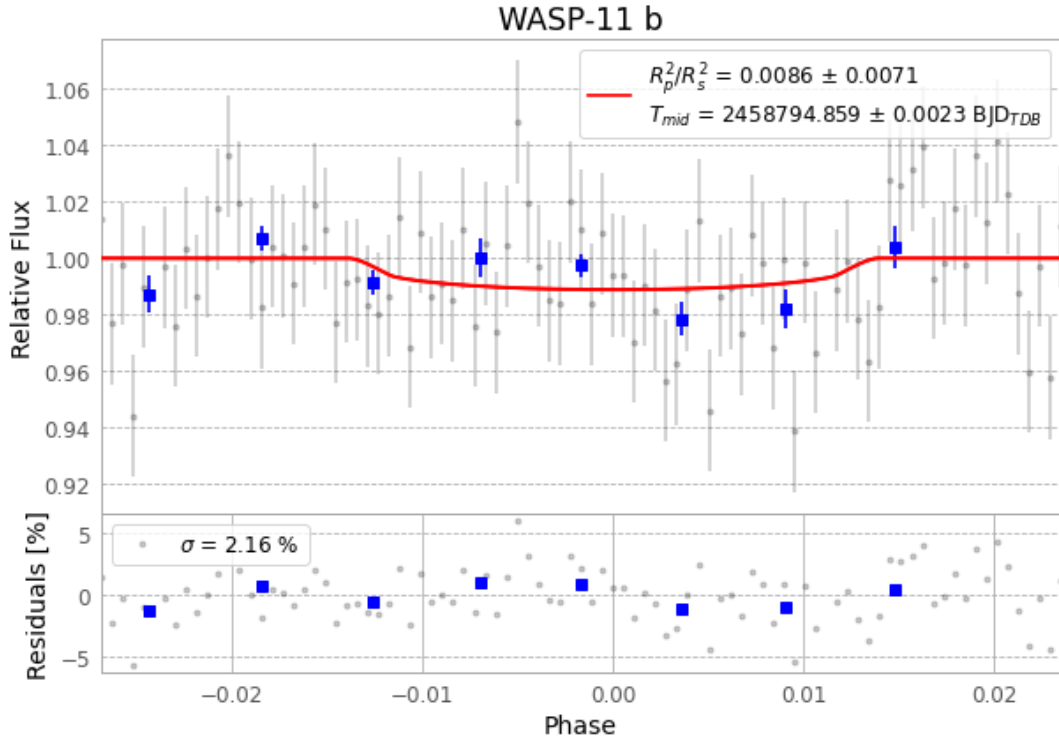


Figure 1 — FinalLightCurve_WASP-11 b_2019-11-06.png (SHA-256 69e121519b0ded22...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 392], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d45a404916a6c023...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), HATP-10191107060312.FITS → HATP-10191107103312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-191107.log (ff678cdfcaaf...), FinalLightCurve_WASP-11 b_2019-11-06.png (69e121519b0d...), FinalLightCurve_WASP-11 b_2019-11-06.csv (78259a9b5591...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 04:34Z.